

TEST PLAN

OrangeHRM Open Source — QA Testing Project

Project Name	OrangeHRM Open Source (Demo Application)
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Role	Software Quality Assurance Engineer
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1. Introduction

This Test Plan describes the testing approach, scope, and deliverables for the OrangeHRM Open Source testing project. The project was carried out independently as a hands-on QA portfolio exercise, with the goal of practicing real-world testing skills against a fully functional HR application rather than a simplified sample project.

The plan covers manual functional testing, API testing, and test automation, and is intended to demonstrate a practical, end-to-end QA workflow: from writing test cases and executing them, to logging defects and building supporting documentation such as a Requirements Traceability Matrix (RTM).

2. Project Overview

OrangeHRM is an open-source Human Resource Management System (HRMS) that supports managing employee records, leave requests, recruitment activity, and user administration. Its public demo instance was used as the Application Under Test (AUT) for this project, since it closely resembles the kind of system a QA engineer would encounter on a real assignment.

Testing was performed using a mix of manual testing for UI and functional flows, Postman for API-level checks, and Selenium WebDriver for automating a core set of regression scenarios.

3. Testing Objectives

- Verify that the core HR workflows behave as expected from a user's perspective.
- Identify, reproduce, and clearly document defects with enough detail for a developer to act on them.
- Validate the behavior and response handling of key REST API endpoints.
- Automate a small set of repeatable, high-value regression scenarios.
- Maintain traceability between requirements and test cases through an RTM.
- Produce documentation (test cases, bug reports, summary report) that follows practices used in real QA teams.

4. Scope of Testing

In Scope

- Authentication Module — Login, Logout, Forgot Password, input/credential validation
- Dashboard Module — widget display, Quick Launch links
- PIM Module — Add Employee, Edit Employee, Employee Search, Employee Details
- Admin Module — Add User, Edit User, Delete User, Search User, User Role assignment
- Leave Module — Apply Leave, Approve/Reject Leave, Cancel Leave, Leave list filtering
- API Testing — Authentication endpoints and selected Employee/Leave endpoints, covering status codes, response structure, and basic negative scenarios

Out of Scope

- Mobile application testing (no dedicated mobile app for the AUT)
- Performance and load testing
- Penetration / security testing beyond basic input validation checks
- Database-level validation — the project uses the public demo instance, with no direct database access
- Third-party integrations (payroll, SSO, etc.)

5. Testing Types and Approach

Functional Testing

Each in-scope feature was tested against expected behavior using a mix of:

- Positive scenarios (valid inputs, expected workflow)
- Negative scenarios (invalid inputs, missing fields, unauthorized actions)
- Boundary checks where applicable (field length limits, date ranges)

UI Testing

Carried out alongside functional testing, focusing on:

- Field labels, buttons, and form layout
- Validation and error messages
- Navigation between modules

Smoke Testing

A short smoke check was run before each test cycle to confirm the build was stable enough to test, covering Login, Add Employee, Employee Search, Apply Leave, and Logout.

Regression Testing

Re-executed after any fix or environment change to confirm previously working functionality was not affected.

API Testing

Performed using Postman.

- Validation areas: status codes, response body structure, response headers, authorization, basic error handling
- Methods covered: GET, POST, PUT, DELETE
- Requests organized into a Postman collection with environment variables for the base URL and credentials

Automation Testing

A small automation framework was built to cover a few stable, high-value scenarios rather than full coverage, which is typical for a project at this stage.

- Tools: Selenium WebDriver, Java, TestNG, Maven
- Design: Page Object Model (POM) for maintainability
- Reporting: Extent Reports with screenshots captured on failure
- Automated scenarios: Login (valid/invalid), Logout, Add Employee, Employee Search, Apply Leave

6. Test Environment

Hardware

- Personal laptop, 8 GB RAM minimum

Operating System

- Windows 10 / Windows 11

Browsers

- Google Chrome (primary)
- Mozilla Firefox (cross-browser spot checks)

Tools

- Postman — API testing
- Selenium WebDriver, TestNG, Maven — test automation
- GitHub — version control and defect tracking (Issues)
- Microsoft Excel / Google Sheets — RTM and test case management

7. Test Deliverables

Documentation

- Test Plan
- Test Cases
- Requirements Traceability Matrix (RTM)
- Test Summary Report

Defect Management

- Bug Reports (grouped by root cause where multiple test cases point to the same underlying issue)
- Defect log tracked via GitHub Issues

API Testing

- Postman collection and environment file

Automation Testing

- Selenium automation scripts (POM structure)
- Extent Reports with execution screenshots

8. Entry Criteria

- The OrangeHRM demo environment is accessible and stable.
- Test cases for the relevant module have been written and reviewed.
- Required test data (sample employee/user accounts) is available.

9. Exit Criteria

- All planned test cases for the in-scope modules have been executed.
- All identified defects are logged, with critical issues reviewed and explained.
- Smoke test passes on the final test cycle.
- RTM and Test Summary Report are completed.

10. Risks and Assumptions

Risks

Risk	Impact	Mitigation
Public demo environment shared with other users may reset or change data	Inconsistent test results	Re-run affected test cases and note any data-related anomalies in results
Limited environment access (no staging/DB access)	Some checks reduced to UI/API level only	Scope adjusted upfront to exclude database-level validation
Browser-specific rendering differences	Minor UI inconsistencies	Spot-check critical flows in a second browser
Time constraints of a solo, self-paced project	Reduced automation coverage	Prioritize automating the highest-value regression scenarios first

Assumptions

- The OrangeHRM demo instance reflects standard out-of-the-box functionality.

- Testing is performed by a single QA engineer (the author), covering planning, execution, and reporting.
- No formal change requests or sign-off process applies, as this is an independent portfolio project.

11. Roles and Responsibilities

This project was carried out independently. All planning, test design, execution, defect reporting, and documentation were handled by a single person.

Role	Responsibility
QA Engineer (Ali Mohamed)	Test planning, test case design, manual execution, API testing, automation, defect reporting, RTM and summary report

12. Success Metrics

- All planned test cases for in-scope modules executed
- Critical defects clearly documented with reproduction steps and evidence
- Core regression scenarios automated and passing reliably
- RTM shows full traceability between requirements and test cases

13. Project Workflow

- Phase 1 — Test Plan
- Phase 2 — Test scenario identification
- Phase 3 — Test case design
- Phase 4 — Test execution
- Phase 5 — Bug reporting
- Phase 6 — RTM preparation
- Phase 7 — API testing
- Phase 8 — Automation testing
- Phase 9 — Test summary report

Approval

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_ Software Quality Assurance Engineer